Unit 4 · Outcome 3 · Scientific Investigation

U·O3 · KK10 · Science communication conventions — Digital Vector EnviroSci

KK10 VCE Environmental Science · Units 3 & 4 · Exam study summary

> A scientific investigation is only finished when someone else can read it, trust it, and repeat it. Conventions of science communication are the agreed rules that let any scientist — in Mildura or Munich — read your poster and understand it exactly the same way. Get them right and your meaning travels cleanly. Get them wrong and a brilliant investigation can read as careless.

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.

THE SHARED LANGUAGE OF SCIENCE

01 The shared language of science

A scientific investigation is only finished when someone else can read it, trust it, and repeat it. Conventions of science communication are the agreed rules that let any scientist — in Mildura or Munich — read your poster and understand it exactly the same way. Get them right and your meaning travels cleanly. Get them wrong and a brilliant investigation can read as careless.

● **NOTE**

3D not available — the orbiting tokens are decorative; every idea below works without them.

One symbol, one meaning. mg/L is milligrams per litre everywhere — no translation needed.

Standard units and methods let another scientist rebuild your test exactly and check your claim.

SI units let your Wirrim Creek result sit beside an EPA dataset or an IPCC figure without conversion errors.

◆ **NOTE**

The KK10 dot point "Conventions of science communication — scientific terminology and representations, symbols, formulas, standard abbreviations and units of measurement." In the exam this is rarely a whole question on its own — it is the quality layer assessed inside data, calculation and poster questions. Marks quietly leak away when units, notation or terminology are sloppy.

PRECISE TERMINOLOGY BEATS EVERYDAY WORDS

02 Precise terminology beats everyday words

Everyday language is vague on purpose — it leaves room for interpretation. Science wants the opposite: one precise term that names exactly what you mean. Swapping casual phrasing for the correct technical term is the quickest mark-saver on a poster.

◈ **ANALOGY**

Analogy · ordering coffee "A coffee, thanks" gets you anything. "A double-shot flat white, oat milk, 240 mL" gets you the same drink every time. Scientific terminology is the barista's docket version of your meaning — specified so tightly it can't be misread.

CHOOSING & LABELLING THE RIGHT REPRESENTATION

03 Choosing & labelling the right representation

A representation is any non-prose way of showing information — a table, graph, flowchart, food web or labelled diagram. The convention has two halves: pick the form that fits the data, then label it so it stands alone.

Continuous independent variable (distance, time, temperature). Plot IV on the x-axis, DV on the y-axis. Use a line of best fit, not dot-to-dot, for a trend.

Categorical / discrete groups (Site A, B, C). Bars separated by gaps. Compares amounts between named categories.

Tests a relationship between two measured variables (turbidity vs DO). Each point is one paired observation; never join the dots.

Flowchart for a procedure or process; food web for feeding relationships. Arrows show direction — energy flows to the consumer.

> ◆ **NOTE**
>
> Graphing conventions checklist Every graph needs: an informative title (“DV vs IV…”), both axes labelled with the quantity and its unit in brackets — e.g. Dissolved oxygen (mg/L) — a sensible scale starting at a stated value, and a legend if more than one data series. A graph that needs the caption explained has already lost marks.

SYMBOLS, FORMULAS & SUBSCRIPTS

04 Symbols, formulas & subscripts

Chemical symbols and formulas are international shorthand. The subscript tells you how many atoms of the element before it — and the case of every letter carries meaning. CO (carbon monoxide) is not CO_2 (carbon dioxide) and neither is Co (cobalt).

Formula	Reads as	Where it appears in EnviroSci
CO_2	1 carbon, 2 oxygen	Enhanced greenhouse effect; respiration; combustion
CH_4	1 carbon, 4 hydrogen	Methane — high warming potential
N_2O	2 nitrogen, 1 oxygen	Nitrous oxide — agriculture, fertilisers
O_2 / O_3	oxygen gas / ozone	Dissolved oxygen; stratospheric ozone
H_2O	2 hydrogen, 1 oxygen	Water — the hydrosphere

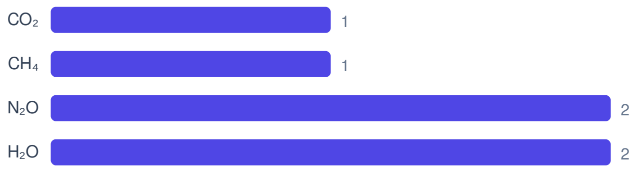


Figure · Formula compared at a glance.

> ◆ **NOTE**
>
> Case & subscript carry meaning CO2 typed flat is tolerated in a rushed logbook, but on a poster write the subscript: CO_2. Capital CH_4 ≠ lowercase. And a coefficient (the big number in front, e.g. 2 H_2O) means “two molecules”, while the subscript means “atoms within one molecule”. Mixing them up changes the chemistry.

Formulas as equations. When you state a formula like Simpson's Index of Diversity, give the equation, define every symbol, and keep variables italic: $SID = 1 - \Sigma(n / N)^2$, where n = individuals of one species and N = total individuals. A formula with undefined symbols is unreadable to a marker.

STANDARD ABBREVIATIONS

05 Standard abbreviations

An abbreviation only works if it is standard and defined on first use. Write the term in full the first time with the abbreviation in brackets — “dissolved oxygen (DO)” — then use DO thereafter. Inventing your own (“H2Oxy”) defeats the purpose.

✓ **TIP**

Exam habit If you use an abbreviation in a short-answer response, define it once. Markers will not assume EC means electrical conductivity rather than, say, the European Commission. One bracketed definition protects every mark that follows.

UNITS OF MEASUREMENT & THE SI

06 Units of measurement & the SI

The International System of Units (SI) is the global standard. A handful of base units are defined first; everything else is a derived unit built by multiplying or dividing them. Reporting in SI is what makes your number comparable to anyone else's.

Base quantity	SI base unit
length	metre (m)
mass	kilogram (kg)
time	second (s)
temperature	kelvin (K)
amount of substance	mole (mol)

Derived quantity	Built from
area	$m^2 = m \times m$
volume	m^3, or litre (L)
concentration	mg/L = mass ÷ volume
density	kg/m^3 = mass ÷ volume
radiative forcing	W/m^2 = power ÷ area

SI PREFIXES & SCALE

07 SI prefixes & scale

Environmental data spans enormous ranges — micrograms of a pollutant up to gigatonnes of carbon. Prefixes rescale a unit by powers of ten so the number stays readable. The quantity never changes — only how you write it.

Prefix	Symbol	Factor	Example in EnviroSci
giga	G	10^9	Gt CO_2 — global emissions
mega	M	10^6	Mha — area of forest cleared
kilo	k	10^3	kW, kJ — energy
milli	m	10^{-3}	mg/L — dissolved oxygen
micro	µ	10^{-6}	µS/cm — conductivity; µg/L — trace pollutant

◆ **NOTE**

The capital-M trap m means milli (one-thousandth); M means mega (one million). They differ by a factor of one billion. Writing Mg/L when you mean mg/L is the single costliest unit slip in environmental data.

NOTATION RULES EXAMINERS NOTICE

08 Notation rules examiners notice

A few small typographic rules separate a polished poster from a rushed one. They feel fussy — but each one removes a possible misreading.

One space between the number and the unit: 25 °C, 8.2 mg/L — not 25°C or 8.2mg/L. (Exceptions: % and pH carry no spaced unit.)

5 kg, never 5 kgs. Unit symbols are invariant — they never take an “s” and never take a full stop (kg, not kg.).

kg, L, K. Units named after a person take a capital symbol but lowercase name — watt → W, kelvin → K. kg ≠ Kg ≠ KG.

Quantities (variables) are italic: T, t, m. Units are upright: K, s, kg. So T = 25 °C.

SIGNIFICANT FIGURES & SCIENTIFIC NOTATION

09 Significant figures & scientific notation

Your numbers must be honest about precision. Significant figures show how finely you actually measured; scientific notation keeps very large or very small numbers compact and unambiguous.

Non-zero digits always count. Zeros between them count. Leading zeros never count (0.0048 → 2 s.f.). Trailing zeros count only with a decimal point (8.20 → 3 s.f.).

A calculated answer can be no more precise than your least precise measurement. If your probe reads to 2 s.f., don't report a mean to 5 s.f. — it invents precision you never had.

◆ **NOTE**

Scientific notation Write a number as a value between 1 and 10 multiplied by a power of ten. 0.00482 → 4.82×10^{-3}; 36 000 000 000 t → 3.6×10^{10} t. The exponent counts how many places the decimal moves: negative for small numbers, positive for large.

COMMAND-WORD DECODER

10 Command-word decoder

◈ **COMMAND WORDS**

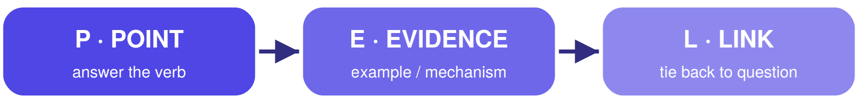
Convention questions often hinge on the verb. Convert wants working and a unit; justify wants a reason. Match your answer's shape to the command word.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

P·E·L WRITING TRAINER

12 P·E·L writing trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✎ **P·E·L WRITING**

Structure a convention answer as Point · Example · Link: name the convention, show it applied (often a corrected version), then link to why it matters for communication. Type below — the meters react to what you write. On her poster Priya writes: “temp was 25°C and DO was 8mg/l”. Identify two conventions she has breached and rewrite the phrase correctly, explaining why each correction aids communication. (4 marks) Point Two conventions are breached: there should be a space between the value and the unit, and the litre symbol must be capital L (and the formula typed as a unit, not casual text). Example Corrected: temperature was 25 °C and dissolved oxygen (DO) was 8.0 mg/L. Link Spacing and correct case remove ambiguity — mg/L cannot be misread as something else — so any reader interprets the measurement exactly as intended, which is the whole purpose of scientific conventions.

WORKED ANSWER · WEAK VS STRONG

13 Worked answer · weak vs strong

Same question, two responses. The difference is entirely in convention discipline.

◆ **NOTE**

Question (3 marks) Priya measured dissolved oxygen at six distances downstream. Identify the most appropriate graph type and justify your choice, referring to the variables. State how the axes should be labelled.

PRACTICE QUESTIONS · 20 MARKS

14 Practice questions · 20 marks

★ **PRACTICE & SELF-MARK**

VCAA-style. Attempt each in your logbook first, then reveal the weak answer, strong answer and marking guide. Mark yourself honestly — the dial tracks your total.

ONE-SCREEN CHEAT SHEET

15 One-screen cheat sheet

- Terminology — one precise technical term, never casual phrasing.
- Representation — line (continuous IV) · bar (categories) · scatter (relationship) · flowchart (process) · food web (feeding). Title + axes with units + legend.
- Symbols / formulas — case & subscript carry meaning: CO_2 ≠ CO ≠ Co. Define every symbol in a formula.
- Abbreviations — define in full on first use: dissolved oxygen (DO).
- SI units — base (m, kg, s, K, mol); derived built from them (mg/L, W/m^2).
- Prefixes — G 10^9 · M 10^6 · k 10^3 · m 10^{-3} · μ 10^{-6}. m ≠ M.
- Notation — space the unit (25 °C); no plurals or full stops (kg); correct case; italic variable, upright unit.
- Precision — report to the s.f. you measured; scientific notation for very large/small (4.82×10^{-3}).

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ **WEAK ANSWER**

Over-precise Priya's probe reads to 0.1 mg/L. She reports the mean DO as 6.47833 mg/L. Claims five-figure precision from a one-decimal instrument — dishonest and mark-losing.

✓ **STRONG / MODEL ANSWER**

Honest She reports the mean DO as 6.5 mg/L (2 s.f.), matching the probe's precision. Precision reported = precision measured. An examiner reads this as careful science.

✗ WEAK ANSWER

A bar graph because it shows the oxygen at each spot. Put distance and oxygen on it." Wrong form (distance is continuous, not categorical), no link to variable type, and axis labelling has no quantities or units. ~0–1 / 3.

✓ STRONG / MODEL ANSWER

A line graph is most appropriate because distance downstream is a continuous independent variable and dissolved oxygen is a continuous dependent variable, so a line can show the trend. The x-axis should read Distance downstream (m) and the y-axis Dissolved oxygen (mg/L), each with its unit in brackets." Correct form, justified by variable type, axes labelled with quantity + unit. 3 / 3.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.